

The following is Section IV, continuing the introduction to the Scalar–Angular Theory (SAT).

I. PATH INTEGRAL QUANTIZATION OF SAT

While the canonical quantization procedure (Section III) provides the local commutation relations necessary for operator formalism, the `**path` integral formulation`**` offers a complementary method crucial for incorporating non-perturbative effects and summing over topological sectors inherent in the Scalar–Angular Theory (SAT).

A. The Functional Integral over Continuous Fields

The partition function Z for a quantum field theory is generally defined as the functional integral over all field configurations $\xi(x)$, weighted by the phase factor determined by the total action S . In the context of the SAT continuous fields $\xi = (\theta_\mu, u_\mu, J^{\mu\nu})$, the general form of the path integral is:

$$Z = \int \mathcal{D}[\xi] e^{iS[\xi]/\hbar}, \quad (1)$$

where $S[\xi]$ is the total action derived from the Lagrangian density $\mathcal{L}_{\text{total}}$. Since the classical dynamics are derived from the variational principle applied to the constrained action, the action S_{SAT} takes the relativistic form:

$$S_{\text{SAT}} = \int d^4x \sqrt{-g} \mathcal{L}_{\text{total}}, \quad (2)$$

where $\mathcal{L}_{\text{total}}$ encapsulates the scalar potential $(\theta_\mu)^2$, the vector kinetic term $(u_\mu)^2$, the filament current energy $(J_\mu^\mu)^2$, the BF topological term $\mathcal{S}_{\text{BF}}$, and the filament kernel $\mathcal{K}_{\text{fil}}$.

B. Hybrid Quantization over Continuous and Discrete Sectors

A distinguishing feature of SAT is its hybrid nature, coupling the continuous field dynamics to a discrete topological twist charge $\tau = (L_{\text{wind}}, L_{\text{link}}, W_{\text{writhe}})$. This requires the partition function to not only integrate over the continuous degrees of freedom but also to `**sum` over all possible topological configurations`**` τ that are stable or admissible within the theory.

The quantization condition is thus a summation over discrete topological invariants combined with a functional integration over the continuous fields, symbolized by ξ :

$$Z_{\text{SAT}} = \sum_{\tau} \int \mathcal{D}[\xi] \exp\left(\frac{i}{\hbar} S_{\text{total}}[\xi, \tau]\right). \quad (3)$$

Here, the total action S_{total} depends explicitly on the discrete charge τ through the topological coupling term $\mathcal{S}_{\text{BF}}$ and the mass-suppressing filament kernel $\mathcal{K}_{\text{fil}}$. The BF action term includes the total twist number $Q(\tau) = L_{\text{wind}} + L_{\text{link}} + W_{\text{writhe}}$ directly.

C. Handling Constraints and the Gravitational Sector

The full functional integral for SAT must address the constraints arising from the time-flow vector u^μ and the gravitational interaction $g_{\mu\nu}$.

1. Constrained Time-Flow Field

The vector field u^μ is dynamically normalized via the constraint $\mathbf{C}_1 : \mathbf{u}^\mu \mathbf{u}_\mu + 1 = 0$, enforced classically by a Lagrange multiplier $\lambda(x)$. In the path integral formulation, this constraint must be handled, typically by integrating out the multiplier λ or choosing a specific gauge fixing for the timelike field u^μ . The associated gravitational Lagrangian $\mathcal{L}_{\text{grav}}$ already includes this multiplier term.

2. Gravitational Constraints

When treating the metric $g_{\mu\nu}$ as a field to be quantized (as required for Quantum Gravity effects), the integration $\mathcal{D}[g]$ must account for the spacetime diffeomorphism invariance. Using the ADM decomposition, the gravitational Hamiltonian density $\mathcal{H}_{\text{GR}+\text{SAT}}$ is expressed in terms of the lapse (N) and shift (N^i) functions and the constraints ($\mathcal{H}_0, \mathcal{H}_i$):

$$\mathcal{H}_{\text{GR}+\text{SAT}} = N\mathcal{H}_0 + N^i\mathcal{H}_i + \lambda(u^\mu u_\mu + 1). \quad (4)$$

These Hamiltonian ($\mathcal{H}_0$) and momentum ($\mathcal{H}_i$) constraints involve the canonical momentum π^{ij} conjugate to the spatial metric h_{ij} .

The full path integral over the combined fields $\mathcal{D}[\xi, g]$ thus requires employing the Faddeev–Popov procedure (or other gauge fixing methods) to properly integrate over the constraints $\mathcal{H}_0$ and $\mathcal{H}_i$, ensuring only the physical degrees of freedom contribute to Z .